



Introduction to Photogrammetry

Lecture 14

Interior Orientation

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In Today's Class

1. Introduction
2. Elements Required for Interior Orientation
3. Coordinate Transformation

1. Introduction

- Interior orientation reconstructs the geometry of the camera at the time of exposure.
- It establishes the relationship between image coordinates and camera coordinates.
- Fiducial marks and camera calibration data are used during this process.
- Interior orientation is the first step in analytical photogrammetric processing.

- Interior orientation restores the original geometry of the aerial photograph.
- It ensures that image measurements are free from geometric inconsistencies.
- The process provides a consistent coordinate framework for further analysis.
- Accurate interior orientation improves the quality of spatial measurements.

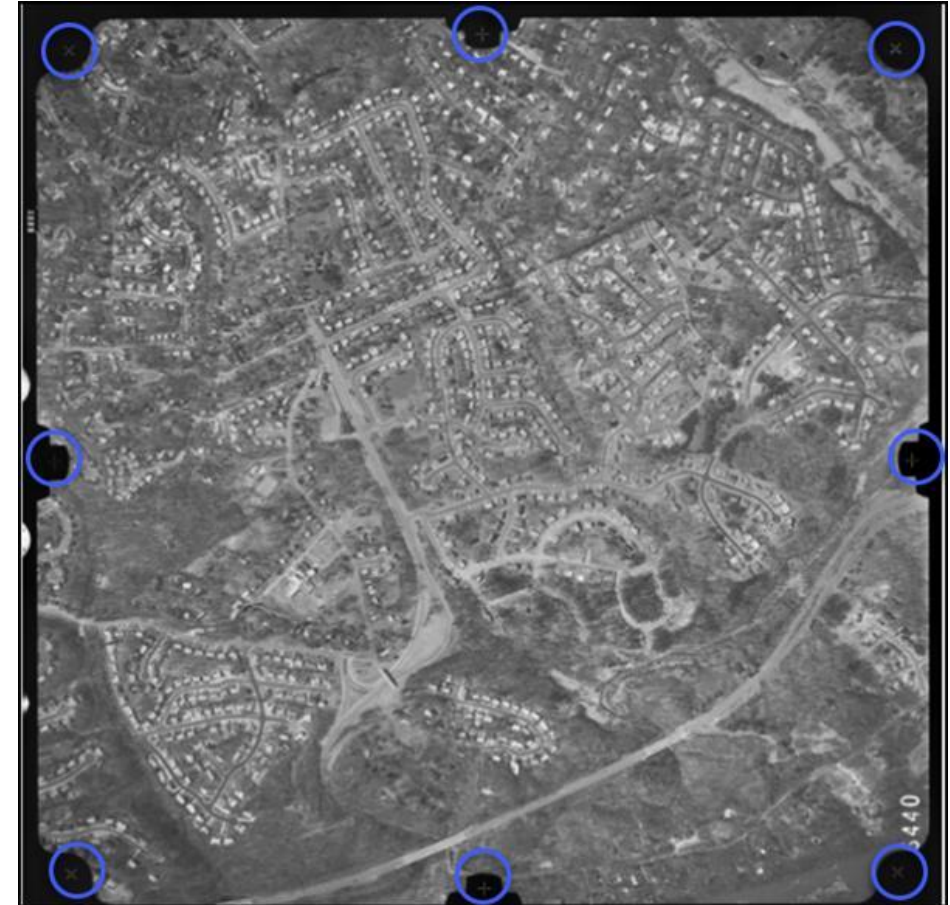
- Interior orientation is used in aerial mapping and topographic surveys.
- The process supports stereo image reconstruction and 3D modeling.
- It is important in digital photogrammetry and remote sensing applications.
- Interior orientation is also used in drone-based mapping systems.

2. Elements Required for Interior Orientation

- Interior orientation requires fiducial marks that are permanently fixed on the photograph.
- Camera calibration data are needed to determine geometric parameters.
- The principal point and focal length must be accurately identified.
- Precise coordinate measurements are necessary for reliable orientation results.

Fiducial Marks

- Fiducial marks are small reference marks located at the edges or corners of aerial photographs.
- These marks define the internal coordinate system of the photograph.
- Fiducial marks remain fixed relative to the camera geometry.
- They provide reference points for transforming image measurements.

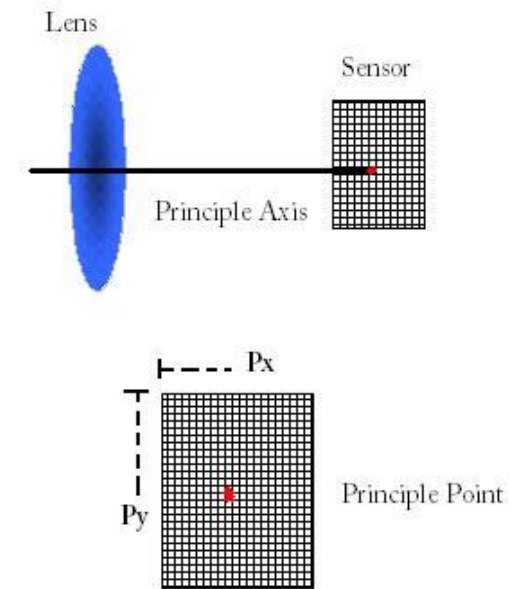


Functions of Fiducial Marks

- Fiducial marks help establish the photo coordinate system accurately.
- They are used to determine the principal point location.
- Fiducial marks support correction of film distortions and scanning shifts.
- Accurate fiducial measurements improve overall photogrammetric precision.

Principal Point

- The principal point is the intersection of the camera's optical axis with the image plane.
- It represents the geometric center of perspective projection.
- The principal point serves as the origin of the image coordinate system.
- Small errors in principal point location can affect orientation accuracy.



Principal Distance

- Principal distance is the distance between the camera perspective center and the image plane.
- It is equivalent to the calibrated focal length of the camera.
- Principal distance controls the scale and geometry of the photograph.
- Accurate focal length information is essential for spatial reconstruction.

Interior Orientation Process

- The interior orientation process transforms measured image coordinates into calibrated photo coordinates.
- Fiducial marks are first measured precisely on the image.
- Transformation equations are then applied to reconstruct camera geometry.
- The resulting coordinate system represents the original camera conditions.

3. Coordinate Transformation

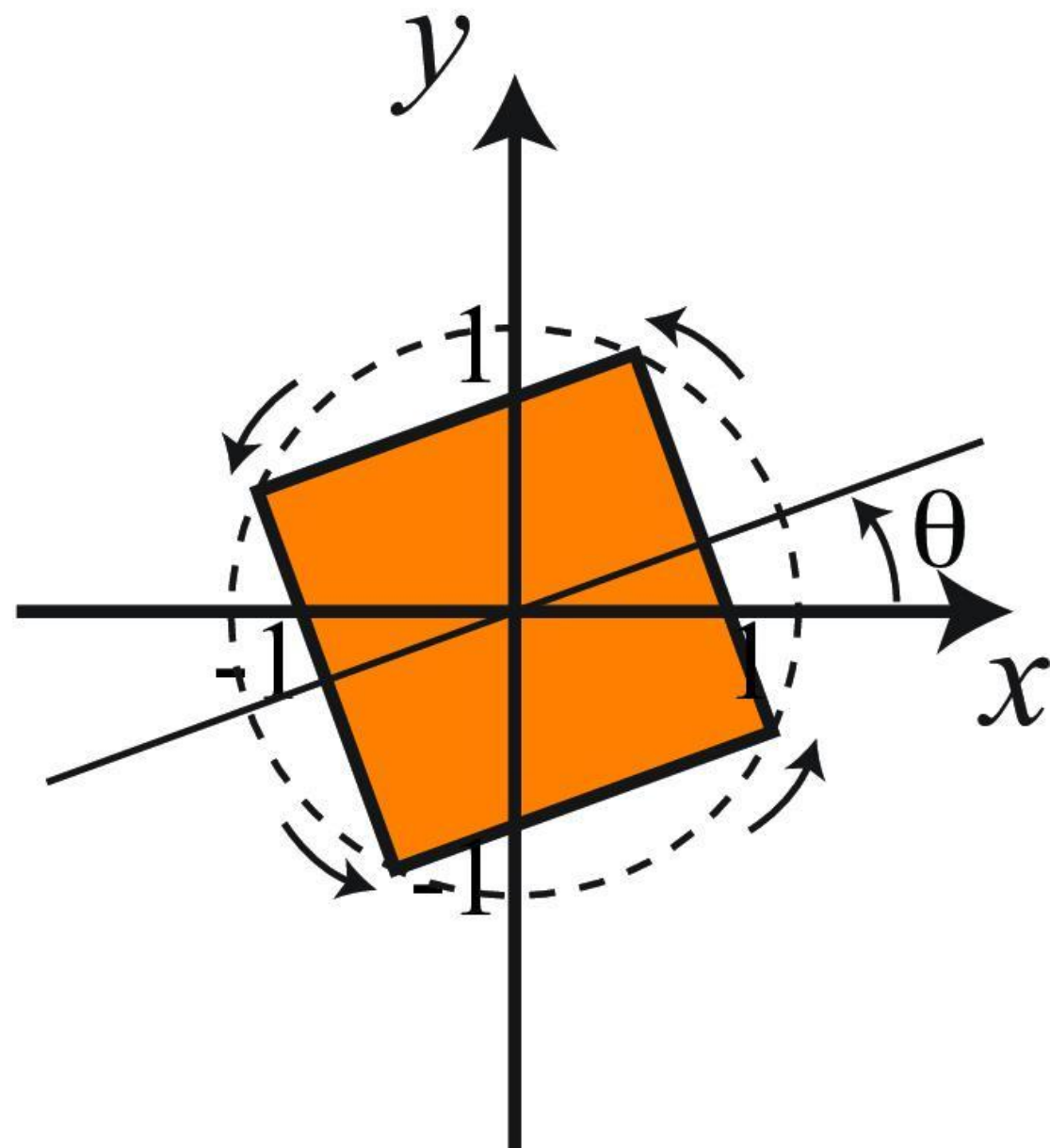
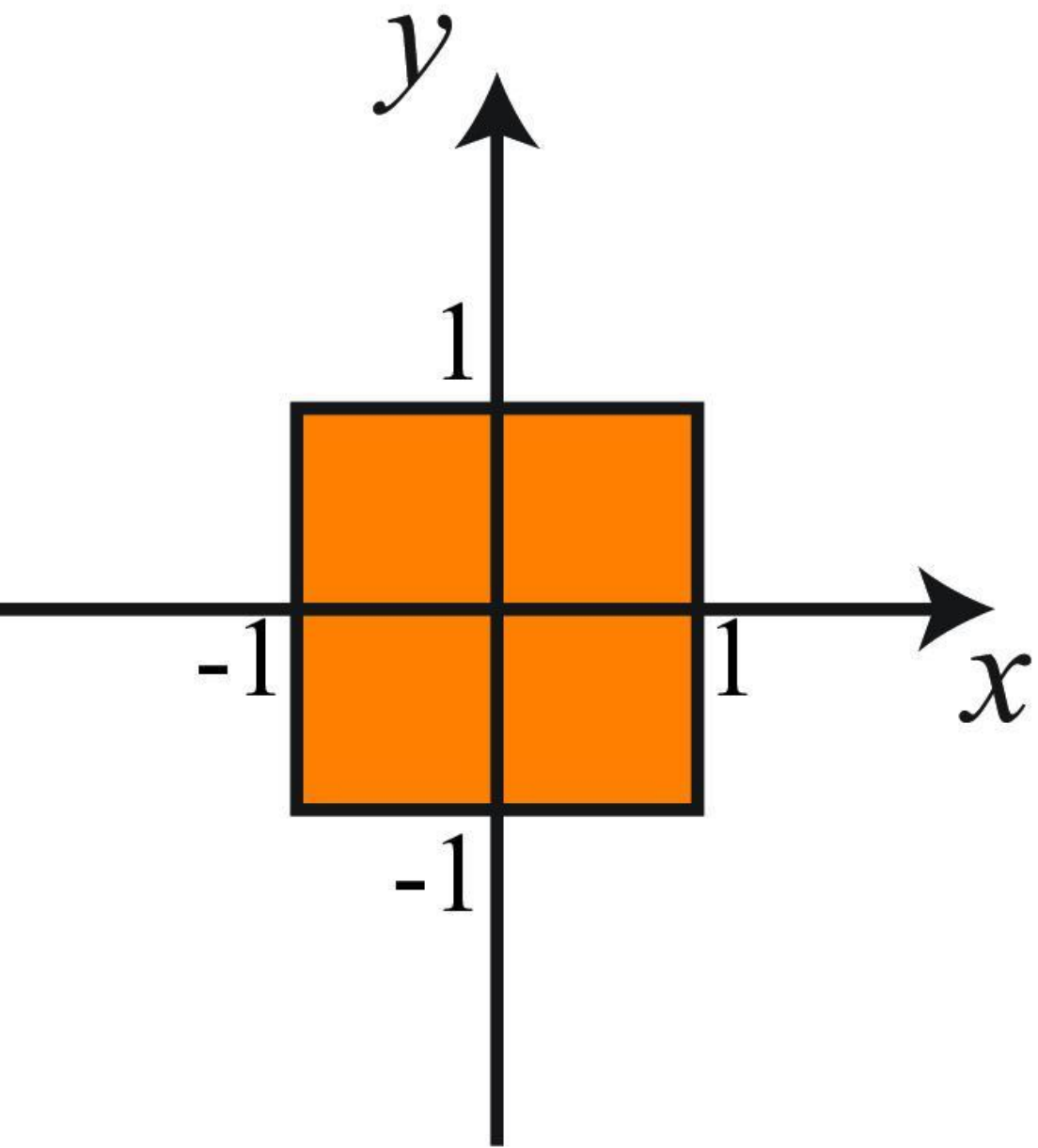
- Coordinate transformation converts coordinates from one reference system to another.
- Transformations are required because image and ground systems are different.
- Mathematical equations are used to perform translation, rotation, and scaling.
- Coordinate transformation is fundamental in analytical photogrammetry.

Translation Transformation

- Translation shifts the origin of a coordinate system from one location to another.
- The shape and orientation of objects remain unchanged during translation.
- Translation equations add or subtract constant values from coordinates.
- This transformation is commonly used during coordinate adjustment procedures.

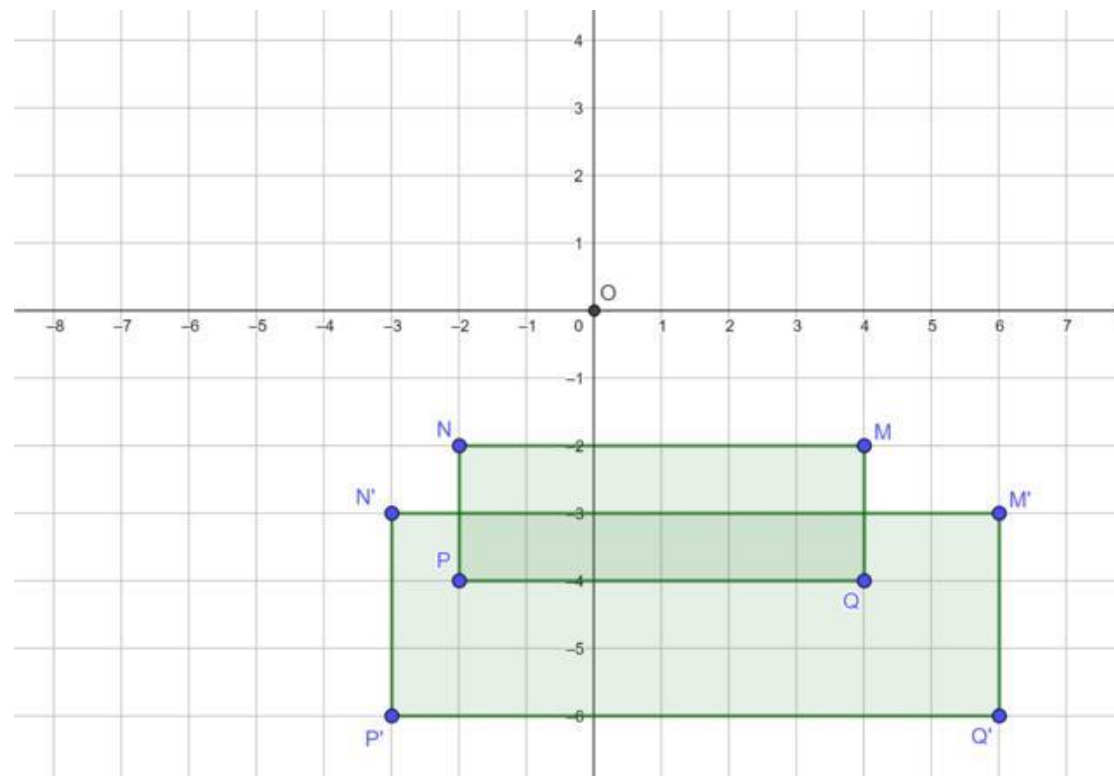
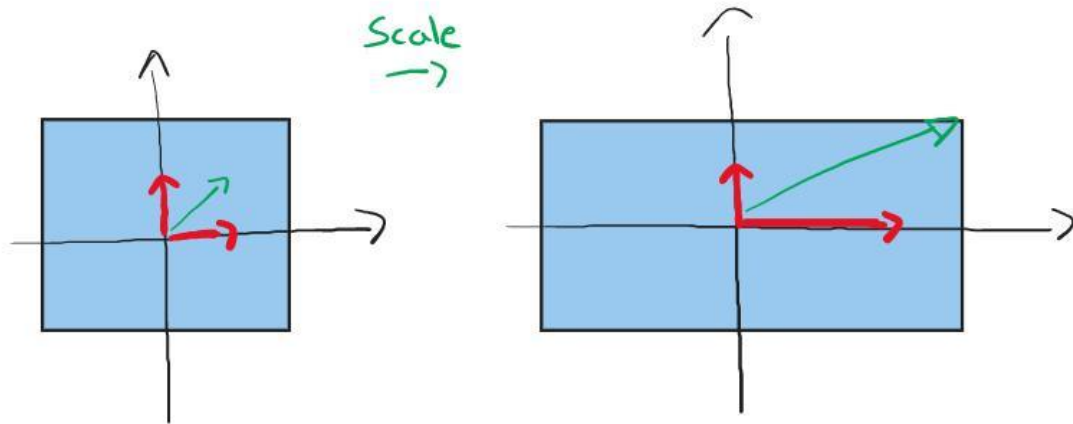
Rotation Transformation

- Rotation changes the orientation of a coordinate system around an axis.
- Rotation angles define the amount of turning between coordinate systems.
- Rotation matrices are used to perform mathematical rotation operations.
- Proper rotation is necessary for aligning photographs correctly.



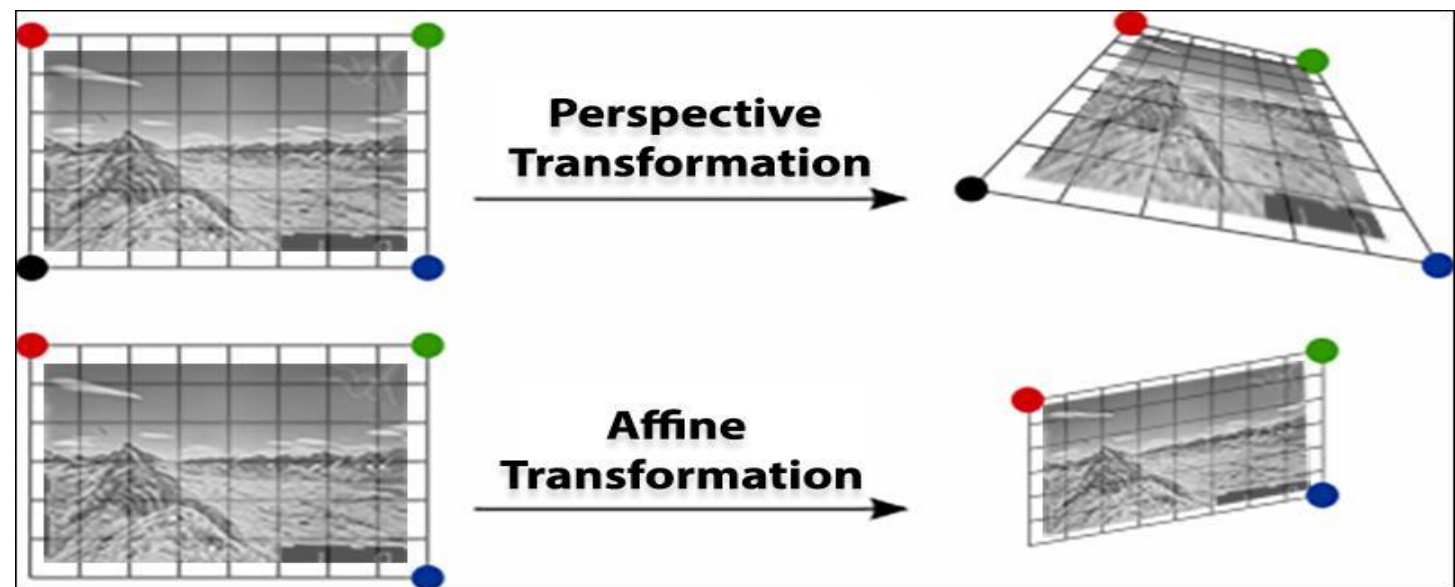
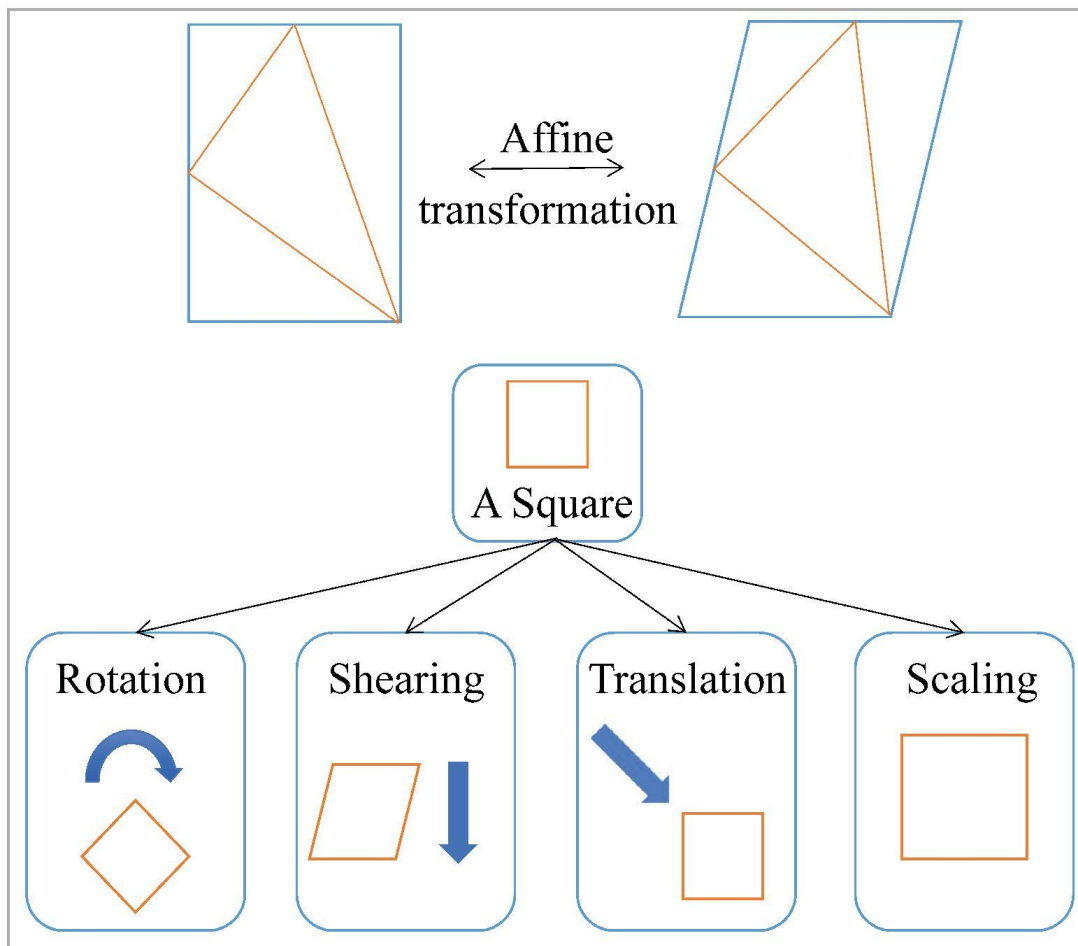
Scale Transformation

- Scale transformation changes the size of coordinate values proportionally.
- Scaling is necessary because image dimensions differ from ground dimensions.
- A scale factor controls the enlargement or reduction process.
- Accurate scaling ensures consistency between image space and object space.



Affine Transformation

- Affine transformation combines translation, rotation, scaling, and skew correction.
- The transformation supports conversion between non-orthogonal coordinate systems.
- Straight lines remain straight after affine transformation.
- Affine models are widely used in photogrammetric coordinate corrections.



Similarity Transformation

- Similarity transformation combines translation, rotation, and scaling operations.
- The transformation preserves the geometric shape of objects.
- It is commonly used in photogrammetric coordinate adjustments.
- Similarity transformation simplifies the conversion between coordinate systems.

Challenges in Coordinate Handling

- **Measurement errors** can affect the accuracy of coordinate transformations.
- **Lens distortions** may introduce geometric inconsistencies in photographs.
- **Inaccurate control points** can reduce the reliability of spatial reconstruction.
- Careful calibration and adjustment procedures are required to minimize errors.



Thank you!

Any Questions
